

AGHA TASHEER SYEDI

Pune, MH, IN | P: +91 79032 26437 | cs.aghasyedi@gmail.com

LinkedIn (<https://www.linkedin.com/in/aghasyedi/>) | GitHub (<https://github.com/aghasyedi>)

Medium (<https://medium.com/@aghasyedi>)

Graduate researcher bridging a strong foundation in Computer Science with advanced expertise in Quantum Mechanics, Quantum Information Theory, and Astrophysics. Experienced in quantum algorithm implementation, post-quantum cryptography, and advanced optics. Passionate about applying computational frameworks to solve complex problems in quantum communication, quantum sensing, and space-based technologies.

SKILLS

- **Quantum Technology & Physics:** Qiskit, Quantum Key Distribution (QKD), BB84 Protocol, Error Correction Codes, Quantum Teleportation, Quantum Metrology (Squeezed States, Atomic Clocks, Ghost Imaging), Photon Statistics.
- **Astrophysics & Space Science:** Cosmological modelling concepts, technical writing on astrophysics.
- **Cryptography & Security:** Post-Quantum Cryptography, Lattice-Based Cryptography, RSA, AES, Hash-Based Signatures, SHA-256.
- **Programming & Classical CS:** Python, Java, C, SQL, JavaScript, Flask, PostgreSQL, Blockchain (Ethereum, Web3.js).

EDUCATION

School of Quantum Technology (SQT), DIAT-DRDO

MTech in *Quantum Computing (Spl: Quantum Communication & Sensing)*

Cumulative CGPA: 8.70 (till 1st Semester)

Focus: Q Communications, Q Computing II, Q Metrology and Sensing, Classical and Post-Quantum Cryptography, Q Optics.

Pune, MH, IN

Expected June 2027

Sam Higginbottom University of Agriculture, Technology and Sciences (SHUATS)

BTech in *Computer Science & Engineering*

Cumulative CGPA: 8.11

Relevant Coursework: Software Engineering; Blockchain Technology; Cryptography

Prayagraj, UP, IN

Aug 2021 - Jun 2025

KEY ACHIEVEMENTS

- **GATE Qualified:** Qualified the Graduate Aptitude Test in Engineering (Computer Science) in both 2024 and 2025.
- **Project Development:** Developed full-stack applications including 'EduPresence' (LMS with Face Recognition) and 'Voyage' (Blockchain-based social platform).

PROJECTS

BB84 Quantum Key Distribution Simulator

Technologies: Vanilla JavaScript, HTML5, CSS3, Canvas API, KaTeX

- Pure client-side interactive web application simulating the end-to-end BB84 QKD protocol without server dependencies.
- Developed real-time visualisations of simulated photon transmission with adjustable channel noise and QBER auditing.
- Implemented advanced cryptographic post-processing pipelines, including Cascade and LDPC error correction, alongside privacy amplification via Universal Toeplitz Hashing to generate unconditionally secure keys.

Planck to Parsec — Astrophysics & Quantum Physics Archive

Authored technical articles and developed a public research archive exploring the intersections of quantum physics and astrophysics.

- Covered advanced theoretical concepts ranging from the Big Bang and cosmological simulations to beam-powered propulsion mechanics.

EduPresence: Learning Management System

Technologies: Python (Flask), PostgreSQL, Face Recognition, Google OAuth, Jitsi Meet

- Built a web-based LMS with specific roles for students, instructors, and admins.
- Designed and implemented a secure, automated attendance system using face recognition and QR-code verification, focusing on authentication accuracy and system reliability.
- Used Google OAuth for login authentication and Jitsi Meet for live video classes.
- Created a dashboard for managing course content and user data.

Voyage: Decentralized Social Media App

Technologies: Blockchain (Ganache), Web3.js, SHA-256, Flask, Python

- Created a social media application using Ethereum smart contracts for data storage.
- Used SHA-256 hashing for user password security.
- Connected the Python backend to a local blockchain network using Web3.js.

EXPERIENCE

LPS Web Technology

Role: PHP & MySQL Intern

Duration: September 2023 - December 2023 (4 months)

- Developed backend functionality using PHP and MySQL, focusing on database management and enhancing backend architecture.

AVingenious Technologies

Role: Web Design Intern

Duration: June 2023 - August 2023 (2 months)

- Designed and implemented responsive web layouts, improving the user experience.

EISystems Technologies

Role: Data Science and Machine Learning Intern

Duration: May 2023 - June 2023 (2 months)

- Assisted in data preprocessing and machine learning model training.

CERTIFICATES

- Completed 'Foundations of Quantum Mechanics' [University of Colorado Boulder] (Coursera, 26 hours) – Nov 2025.
- Completed 'AI & Quantum Computing Mastery' (Udemy, 5.5 hours) – Aug 2025.
- Completed 'Full Stack Development Bootcamp' (upGrad) – December 2024.

POSITIONS OF RESPONSIBILITY

- **Coordinator, Rangoli Event, SPANDAN 2026** — School of Quantum Technology, DIAT
- **Sports Coordinator, School of Quantum Technology, SPANDAN 2026** (Jan 19–27, 2026)

LANGUAGES

Hindi: Native

English: Professional Working Proficiency